AI Study Planner using IBM watsonx.ai and Agentic AI

Problem Statement

Students often face difficulties in planning their studies effectively due to multiple subjects, limited study time, varying difficulty levels, and approaching examination dates. Traditional study schedules are usually static and fail to adapt to individual learning needs, resulting in poor time management, ineffective revision, increased stress, and reduced academic performance.

An intelligent system is required to generate personalized study plans based on each student's academic requirements, available study hours, examination schedule, and learning goals. Such a system should assist students in organizing their studies efficiently while improving productivity and maintaining a balanced learning routine.

Objectives

The main objectives of the AI Study Planner are:

- Develop an AI-powered web application that generates personalized study plans.
- Create daily and weekly study schedules based on student inputs.
- Prioritize subjects according to difficulty level and examination dates.
- Generate revision schedules to improve knowledge retention.
- Provide productivity tips and motivational guidance.
- Demonstrate an Agentic AI approach using multiple logical agents.
- Integrate IBM watsonx.ai foundation models through IBM Cloud.

Proposed Solution

The proposed solution is an AI-powered Study Planner developed using **Python**, **Flask**, and **IBM watsonx.ai**. The application collects information such as student name, subjects, available study hours, examination dates, and academic goals through a web interface.

The backend processes the user input and creates a structured prompt, which is sent to the **Meta Llama 3.3 70B Instruct** foundation model hosted on IBM watsonx.ai. The model generates a personalized study plan containing daily and weekly schedules, subject prioritization, revision planning, productivity tips, and motivational guidance.

The application follows an **Agentic AI** architecture by organizing the workflow into logical agents such as Student Profile Agent, Planning Agent, Subject Priority Agent, Revision Agent, and Motivation Agent.

Technologies Used

IBM Technologies

- IBM Cloud
- IBM watsonx.ai
- IBM Watson Studio
- Meta Llama 3.3 70B Instruct Foundation Model

Programming & Development

- Python
- Flask
- HTML5
- CSS3
- Bootstrap 5
- JavaScript
- IBM watsonx.ai Python SDK

Development Tools

- Visual Studio Code
- Git
- GitHub

System Workflow

1. Student enters study details.
2. Flask receives and validates the input.
3. Agent Controller prepares the prompt.
4. IBM watsonx.ai processes the prompt using the Meta Llama 3.3 foundation model.
5. AI generates a personalized study plan.
6. The study plan is displayed on the web application.

Expected Outcome

The AI Study Planner provides students with personalized study schedules that improve time management, subject prioritization, revision planning, and learning efficiency. The application demonstrates the practical use of IBM watsonx.ai and Agentic AI concepts while offering an intelligent, user-friendly, and scalable educational solution.

Conclusion

The AI Study Planner successfully integrates IBM watsonx.ai with a Flask-based web application to generate intelligent and personalized study plans. By utilizing an Agentic AI approach and a foundation model, the system helps students organize their studies effectively, improve productivity, and prepare efficiently for examinations.